Notes from Introduction to group theory

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1 Lesson 1

Definition 1.1. A group is a set coupled with an internal product $\cdot$:

$$\cdot : G \times G \to G \tag{1}$$
$$(g_1, g_2) \to g_1 g_2 \tag{2}$$

such that:

$$\text{A1:}(g_1 g_2) g_3 = g_1 (g_2 g_3) \tag{3}$$
$$\text{A2:}\exists e : \forall g, ge = eg = g \tag{4}$$
$$\text{A3:}\forall g, \exists g^{-1} : gg^{-1} = g^{-1} g = e \tag{5}$$

Definition 1.2. A group is called abelian iff $\forall g, k \in G, gk = kg$

One can prove the **uniqueness of the identity element**, by proof of contraddiption. Assume $e, f \in G$ are both the identity element with $f \neq e$, then $ef = f$ by A2, but also $ef = e$, which proves the claim.
One can also prove the **uniqueness of the inverse element**, again by contraddiption. Given $g, a, b \in G, a \neq b$, assume that both a, b are the inverse element of g. Then, by A1 and A3:

$$ga = gb \implies (ag)a = (ag)b \implies a = b$$

wich is a contraddiption of the hypothesis, proving the theorem.

1.1 Algebraic structures

There exists other algebraic structures with less proprierties of a group, we only cite them:

- Magmas: Set with closed operation $\cdot : M \times M \to M$
- Semigroup: Magma + associativity
- Monoid: Semigoup + existence of the identity element

Claim 1.3 (direct product)**.** *Given two groups $(G, \times_1), (L, \times_2)$ the* **direct product** *$P = G \times L$ of the two groups is still a group, with respect to the product defined by:*

$$\cdot : P \times P \to P$$
$$((g_1, l_1), (g_2, l_2)) \to (g_1 \times_1 g_1, l_1 \times_2 l_2)$$

Example 1.4. two-element group The first non-tivial group (wich is $\{e\}$) is the group $G = \{e, a\}$.
One can show that the only way to make a group with two elemets, is to define the following products:

$$ee = e \quad ea = a \quad ae = a \quad aa = e$$

one cannot define $aa = a$ because a must have an inverse element.

To represent the operation of $\cdot$ on the elements of the group we make use of the **Cayley table**. Using the Cayley table, we can represent the group of the leading example as:

	e	a
e	$ee = e$	$ea = a$
a	$ae = a$	$aa = e$

Definition 1.5 (cardinality)**.** The caridnality of a group $|G|$ is the number of its elements.

Groups can be finite, infinte and discrete, or infinite and continuos

Example 1.6. some examples of infinte groups are $(\mathbb{Z}, +), (\mathbb{R}, +), (\mathbb{R}/\{0\}, \times), (\mathbb{R}_+, \times)$

Definition 1.7 (General linear group)**.** we define $GL(n) \coloneqq \{(n \times n)\text{matrices with} \det \neq 0\}$ the General linear group, with respect to matrix multiplication

2 Lesson 2

Definition 2.1 (Dihedral groups)**.** We define Dihedral groups $D_n \coloneqq$\{symmetries preserving the shape of a regular polygon with n nodes\} , with respect to the composition as the product of the group.

Example 2.2 (D_2)**.** D_3 is the group of symmetries of the triangle. We have 3 rotations and 3 reflections, making $|D_3| = 6$.
D_1 is a peculiar example, because $D_1 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (see this)

In general, $|D_n| = 2n$

2.1 Orthogonal groups

Orthogonal groups are groups that preserve the distance in an Euclidian space[1]. In coordinates, Orthogonal groups are groups that preserve the orthogonality of the reference frame.

Definition 2.3 (Orthogonal Group)**.** We define

$$O(n, \mathbb{R}) \coloneqq \{M \in GL(n) \text{ such that } M^T M = M M^T = \mathbb{1}\}$$

$O(n, \mathbb{R})$ is the group of all the isometries of $\mathbb{R}^n$

Question 2.1. What does $MM^T = \mathbb{1}$ have to do with orthogonality? given the map $m : V \to V$, we ask m to be an isometry with:

$$v^i v_i = |v|^2 = |Mv|^2 = (Mv)_i (Mv)^i$$

MISSING!!:Lie groups.

2.2 More definitions

Definition 2.4. Given $(G, \cdot)$, a subset $H \subset G$, H is a **subgroup** iff $(H, \cdot)$ is a group.
Every group has two **improper subgroup**: $\{e\}, G$; all groups that are not improper are **proper**.
A subgroup H is defined **normal** iff $\forall h \in H, \forall g \in g, ghg^{-1} \in H$.

Definition 2.5. A group G is defined:

- **semisimple** iff none of its proper groups are normal and abelian.
- **simple** iff none of its proper groups are normal.

[1] space with posdef metric

N.B.: Being semisimple is more stringent than being normal, for a group G with only one (proper) normal and non-Abelian subgroup is semisimple, but not simple.

Definition 2.6 (Group homomorphism)**.** Given $(G_1, \cdot), (G_2, \times)$, a map $\rho : G_1 \to G_2$ is a **group homomorphism** iff:

$$\forall g, l \in G_1, \rho(g \cdot l) = \rho(g) \times \rho(l)$$

if the map ρ is a **bijection**, then ρ is an **Isomorphism**. If there exists an isomorphism between two groups, they are called **isomorphic**

Claim 2.7. *In the set of all groups, being isomorphic is an equivalence relation*

Definition 2.8 (Generating set of a grop)**.** Given the group G and the subset $S \in G$, the group $\langle S \rangle$ is the smallest subgroup of G containing all elements of S.
Its elements are called **letters**, and a generic product is called a **word**.

Definition 2.9 (Generators of a group)**.** Given a group G, the smallest subset $S = s_1, s_2, \ldots$, $S \subset G$ such that:

$$\forall g \in G,\ g = s_{j_1}^{i_1} s_{j_2}^{i_2} s_{j_3}^{i_3} \ldots$$

is called **generator of the group G**, and its elements are called generators.
In simple words, the smallest S such that $\langle S \rangle = G$ is a generator of G.

for some generating set S, there can be products such that

$$r = s_{j_1}^{i_1} s_{j_2}^{i_2} s_{j_3}^{i_3} \cdots = e$$

Such product is called a **relation**

Definition 2.10 (Presentation of a group)**.** A presentation of a group is a way to describe a group by describing its **generators** and its **relations**.
For example, $\mathbb{Z}_n = \langle a | a^n \rangle$

3 Examples of groups

3.1 Cyclic groups

Definition 3.1 (Cyclic groups)**.** As seen above, abstract cycle groups are groups with only one generator.

Rapresentnation of cyclic groups include:

1. $\mathbb{Z}_n = \{1, a, a^2, a^3, \ldots a^{n-1}\}$ with respect to the product as group product
2. $\mathbb{Z}_n = \mathbb{N}/(\cong (\text{mod n})) = 0, 1, 2 \ldots, n-1$ with respect to the addition modulo n
3. Rotational symmetries on n-regular polygon
4. $\mathbb{Z}_n = \{e^{i2k\pi/n}\}$ the points on the unit complex circle, with $\cdot$ as the product composition.

for $n = 1, 2, 3, 5$ the only groups with $|G| = n$ are $\mathbb{Z}_n$. The only exception is for $n = 4$, as $\mathbb{Z}_2 \times \mathbb{Z}_2 \ncong \mathbb{Z}_4$. In fact, it turns out that:

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \langle (e, a), (a, e) | (e, a)^2, (a, e)^2 \rangle$$

3.2 Dihedral group

The two groups of order 6 are $\mathbb{Z}_6$ and D_3, the group of the symmetries of the triangle.
D_3 can be represented as $D_3 = \{e, a, b, aba, ab, ba\}$. The Cayley table is:

$\cdot$	e	a	b	aba	ab	ba
e	e	a	b	aba	ab	ba
a	a	e	ab	ba	b	aba
b	b	ba	e	ab	aba	a
aba	aba	ab	ba	e	a	b
ab	ab	aba	a	b	ba	e
ba	ba	b	aba	a	e	ab

Where a, b, aba are the reflections, and e, ab, ba are the rotations of $0, 120, 240$ degrees. The table can be built nothing that $(ab)^3 = e$. In fact, for D_3 one can notice that:

$$D_3 = \langle a, b | a^2, b^2, (ab)^3, (ba)^3 \rangle$$

These properties allow the computation of some product of the table. For example, one can compute $b(aba)$ (row 3, col 4) by:

$$ababab = e \implies (aa)baba(bb) = ab \implies abab = ba$$

3.3 Permutations

Definition 3.2. Given the set X, $X \neq \{\emptyset\}$, a **permutation** is a bijective map:

$$f : X \to X$$

The set of all permutations is called **Symmetric group**

$$S_n = \mathrm{Perm}(X) = \{f : X \to X \text{bijective}\}$$

and it forms a group with respect to map composition

Some permutations we already encountered are $S_1 = \{e\}$, $S_2 = \{e, a\} = \mathbb{Z}_2$, $S_3 = D_3$.

3.3.1 Representing permutations

The first way to represent a permutation is by the **two-lines** notation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \dots & \sigma(n) \end{pmatrix}$$

The first line contains the elements of the set, the second line contains what the elements are mapped to. This type of representation makes it easy to combine permutations. For example, if

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$$

Then AB is calculated right to left as $1 \to 2 \to 1, \quad 2 \to 4 \to 4 \quad 3 \to 3 \to 2 \quad 4 \to 1 \to 4$, making

$$AB = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 2 & 3 \end{pmatrix}$$

A more compact way to represent a permutation is by **Cyclic notation**, where, for example, A would be written as:

$$A = (1, 3, 2)(4) = (1, 3, 2) = (2, 1, 3) = (3, 2, 1)$$

This notation is particularly effective because the inverse $A^{-1} = (1, 2, 3)$ is found simply by inverting the order.
It is also notable that every permutation can be decomposed in twp-elements cycle, for example $A = (1, 3)(3, 2)$.

Definition 3.3. For every permutation P the **Sign of the permutation** is defined as:

$$\text{sng(P)} = \begin{cases} +1 & \text{if P is written as a product of } \textbf{even} \text{ two-terms cycles} \\ -1 & \text{if P is written as product of } \textbf{odd} \text{ two-term cycles} \end{cases}$$

Theorem 3.4 (Cayley's theorem)**.** *Every group of order N is isomorphic to some subgroup of S_n (equivalently, homomorphic to S_n)*

Proof. Let G be a group with $|G| = N$.
Let's consider $K = \{\text{functions} f_g : x \to g \cdot x\}$ the set of the left actions of G onto itself. Of course, K is a group with respect to function composition. These functions are **bijections**, because $\forall f_g \in K, \exists f_g^{-1} = f_g^{-1}$. This proves that $K \subseteq S_N$.
Now it is sufficient to verify that the map $\rho : (G, \cdot) \to (K, \circ) \subset S_N$ which is a group homomorphism to prove the theorem □

3.4 Partitions and young diagrams

Definition 3.5. A cycle is called "reduced" if every element of the set only appears once.
For example:

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 6 & 4 & 1 & 5 & 2 \end{pmatrix} \quad \Longrightarrow \quad A = (1,3,4)(2,6)(5)$$

Claim 3.6. *For a "reduced" cycle, if n_i is the length of the $i-$th cycle, it is obvious that:*

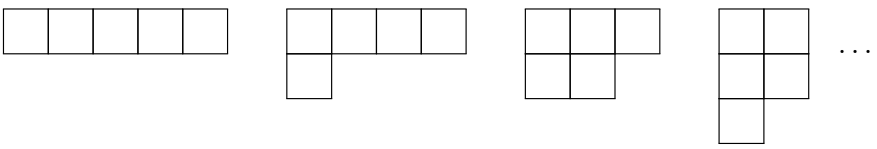
$$\sum n_i = N$$

*hence $\{n_1, n_2, \dots\}$ is a **partition** of N.*

For a generic n, the number of its partitions is $p(n)$, which is found by the generating function:

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{n=1}^{\infty} \frac{1}{1-x^n}$$

An interesting way to represent partitions to use **Young Diagrams**; just like *regoli*, if $N = 5$, the partitions of N are:



3.5 Braid group

Definition 3.7. Given n strands in a 2-dimensional space, the **Braid group** B_n is the set whose elements are all the possible configurations of strands (a more accurate definitions is found by equivalence class from ambient isotopy). The operation of this group is the "braiding" of the strands.

Some notable examples of braid groups are:

$$\begin{cases} B_1 \cong \mathbb{Z}_1 = e \\ B_2 \cong (\mathbb{Z}, +). \end{cases}$$

The last congruence can be seen as follows: braiding two strands on a plane can be seen as twisting a point around a fixed spot. Every twist of the strands can be seen as equivalent of a loop[2](under the equivalence classes of homotopy) on a pierced plane, making the braid group equivalent to the fundamental group of the pierced plane, which is $\mathbb{Z}$.

[2]Basically, a loop is a function $f; [0,1] \to M$ such that $f(0) = f(1)$

3.5.1 Braid group generators and relations

The generic element $\sigma \in B_n$ can be seen as the product:

$$\sigma = \sigma_{j_i}^{i_1} \sigma_{j_2}^{i_2} \dots$$

where the generators $\{\sigma_k\}_{k=1}^{n-1}$ represent the twist of the k strand in front of the $k+1$ strand. The inverse element of σ_k is σ_k^{-1}, which is the opposite twist of the strings.
The **Reidemeister move** of 2nd type is simply $\sigma_k \sigma_k^{-1}$ which is the first relation.
One can also claim that:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \qquad \text{if} \quad |i-j| > 1$$

Claim 3.8. *Given 3 consequential strands, it is true that:*

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_1 \sigma_{i+1}$$

These relations give us the presentation of the group:

$$B_n = \langle \sigma_1, \sigma_2, \dots \sigma_{n-1} | \sigma_i \sigma_j = \sigma_j \sigma_i,\ \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_1 \sigma_{i+1} \rangle$$

4 Continuous groups

Definition 4.1. A group is said to be **continuous** iff its set contains an uncountable number of elements.(e.g.$\mathbb{Z}, \mathbb{Q}$ are not continuous)

In general, we must characterize the *dimension* of G.

Definition 4.2. the **dimension** of a group is the minimum number of independent real parameters needed to parametrize all of its elements. These real parameters are called **coordinates**.[3]

For example, $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. Because of the constraint, $\dim(U(1)) = 1$.

Definition 4.3. Given $g, g' \in G$ continuous groups, if G has a structure of a *manifold*, and the operation of the group is such that $g'' = g'g$, $\forall g, g'$ it is granted that g'' depends **smoothly** on the coordinates of g, g'', then G is called a **Lie group**. As physicist, we are interested in smooth objects, and we will work mainly on Lie groups.

Some examples of groups are:

$$(\mathbb{R}, +) \qquad (\mathbb{R}^n, +) \qquad (\mathbb{C}, +) \qquad U(1), \qquad U(1)^n = U(1) \otimes^n \qquad (\text{Mat } (n \times n) \ , +) \simeq \mathbb{R}^{n \times m}$$

The fact that U(1) is not homomorphic to $\mathbb{R}$ even tho they have the same dimension, is related to the fact that they have different relations. Also, they don't have a smooth bijection because they are not diffeomophic because they have two different fundamental groups.
Also, U^n is isomorphic to an $n-$dimensional torus.
Examples

4.1 Examples of continuous (Lie) groups

Definition 4.4. The **General linear transformations**$GL(n, \mathbb{R})$ group is:

$$GL(n, \mathbb{R}) \coloneqq \{A \in \text{Mat}(n, \mathbb{R}), \det(A) \neq 0\}$$

all matrix group must subgroup of GL, to garante an inverse element.

[3] this is similar to the manifold definition)

Claim 4.5. *The dimension of $GL(n;\mathbb{R})$ is n^2, because the condition is true only in a sub-manifold of Matr($n \times n$).*

One can also define $GL(n,\mathbb{C}) \coloneqq \{A \in \mathrm{Mat}(n,\mathbb{R}), \det(A) \neq 0\}$ and note that $GL(\mathbb{R}) \subset GL(\mathbb{C})$

Definition 4.6. The **Special linear transformation** group is defined as:

$$SL(n,\mathbb{R}) = \{A \in \mathrm{Mat}(n,\mathbb{R}) = \det(A) = 1\}$$

this group is a proper subgroup of $GL(n,\mathbb{R})$. Of course, $\dim A = n^2 - 1$

Definition 4.7. The **Orthogonal** group is:

$$O(n) = \{A \in \mathrm{Mat}(n,\mathbb{R}) : A^T = A^{-1} \iff AA^T = A^T A = 1\}$$

which has $\dim(O(n) = n(n-1)$. This is also the group of isometries of $\mathbb{R}^n$, because $v^2 = (Mv)^2 \implies v^T M^T M v = v^T v$. Taking the determinant of the second equation shows that $\det\{A\} = \pm 1$.
The **Special linear group** $SO(n,\mathbb{R})$ is $O(n)$ with $\det = 1$

Definition 4.8. The **Unitary group** of degree n is:

$$U(n) = \{A \in \mathrm{Mat}(n,\mathbb{C}) : A^{-1}A^{+}\}$$

The unitary group is of special interest because it's the group of transformations on $\mathbb{C}^n$ that preserves the complex amplitudes; if $u \in \mathbb{C}^n$, then:

$$u^2 = (Au)^2 \iff (Au) \dagger (Au) = u \dagger u \iff A \dagger A = 1$$

We can calculate the dimension of $U(n)$ by: $\dim(U(n)) = 2n^2 - 2\frac{n(n-1)}{2} = n^2$

Definition 4.9. We define the **special unitary group** as the group:

$$SU(n) \coloneqq \{M \in U(n) : \det\{A\} = 1\}$$

obviously, $\dim SU(n) = n^2 - 1$, and $SU(n) \subset U(n)$

Note: If we consider the following mappings $i : SU(n) \to U(n)$ and the $\det : SU(n) \to U(n) : M \to \det\{M\} = e^{i\alpha}$, we can see that

$$1 \to SU(n) \xrightarrow{i} U(n) \xrightarrow{\det} U(1) \to 1$$

is an **exact sequence**, as $\ker \det = \mathrm{Im}\, i$

Definition 4.10. We define the **symplectic group**:

$$Sp(n) \coloneqq \{A \in GL(2n) : A^T \Omega A = \Omega\}$$

where Ω is the symplectic form.

4.2 Definitions in continuous groups

Definition 4.11. Considering the group G, the set $X \neq \{\emptyset\}$, a **left action** L of G on X is a group homomorphism:

$$L : G \to \mathrm{Perm}(X) : g \to L_g$$

such that $\forall g_1, g_2 \in G, L(g_1 g_2) = L_{g_1 g_2} = L_{g_1} \circ L_{g_2}$. If $\exists L$, then X is called **left group-space**

Obs: the left group action may also be defined by:

$$L : G \times X \to X : (g,x) \to L_g(x)$$

We define **right action** in the same way, with $G_{g'g} = R_g R_{g'}$.

Definition 4.12. The **orbit** of an element $x \in X$ is the set:

$$O_x = \{x' \in X : \exists g \in G \text{such that} x' = L_g(x)\}.$$

In other words, the orbit of a point is the set of all points that can reach x via the left action of G

For example, if $X = \mathbb{R}^2$, and $x \in X$, $O_x = \{p \in C_{|p|}(0)\}$

Claim 4.13. *Belonging to the same orbit is an equivalence relation.*

Proof. We must show reflexivity, symmetry and transitivity.
1. $L(e) : e \to L_e = \text{Id}$ on X, thus $x \in O_x$.
2. If $x' \in O_x$, then $\exists L_g : L_g(x) = x'$. But, being $L_g \in \text{Perm}(X)$, $\exists L_{g^{-1}}$ such that $L_{g^{-1}}(x') = x$. Thus, $x \in O_{x'}$
3. If $x' \in O_x$, $x'' \in O_{x'}$, then $x \in O_{x''}$. We prove this by composing maps. □

Definition 4.14. Given a left action, it is called to be **transitive** if all elements of x belong to the same orbit.

For example, if we consider $X = \mathbb{R}^2$, with respect to the translations on the plane $(\mathbb{R}^2, +)$, the left action of $\mathbb{R}^2$ on $\mathbb{R}^2$ is transitive.

Definition 4.15. We define the **Isotropy group** (stability group / stabilizer / little group) of $x \in X$, the subgroup of G whose elements (when L acts on them) leave x invariant.

$$I_x = (g \in G : L_g(x) = x)$$

if $I_x = G$, then x is called a **fixed point** of the action of L.

Claim 4.16. *$I_x \subset Perm(X)$ is always a subgroup of the action of the group.*

Proof. By definition, if $g, g' \in I_x$ then $L_g(x) = L_{g'}(x) = x$. So, the composition

$$L_{gg'}(x) = L_g \circ L_{g'}(x) = x.$$

is still in the isotropy group. Furthermore, $L_e \in I_x$ because $Id(x) = x$. Finally, $I_{g^{-1}} \in I_x$ because:

$$E(x) = (L_g)^{-1} \circ L_g(x) = L_{g^{-1}}(x) = x$$

□

If we consider $X = G$, we have two cases:
1. The **translations**, $L : G \to \text{Perm}(G) : g \to L_g$, with $L_g : x \to gx$. The translation is always transitive
2. The **conjugation**, which has action $L_g : x \to gxg^{-1}$.